

Zubair Mughal

LIVE CODING CLASSES FOR KIDS, TEENS & BEGINNERS

JS

COURSE 02 OF 12
CURRICULUM GUIDE

JavaScript for Kids & Teens

Interactive programming for the browser

GRADE BAND

Grade 8–12

DIFFICULTY

Beginner

DURATION

12 weeks · 2 sessions/week

ABOUT THIS COURSE

Course Overview

JavaScript is the language that makes the web move, react and respond. This course takes students from their first script to a genuinely interactive, deployed browser application.

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Prerequisites: Basic programming exposure recommended (e.g. Python for Kids & Teens), but not required.

LEARNING OUTCOMES

What Students Will Learn

- Understand how JavaScript fits alongside HTML and CSS on a page
- Use variables, functions and control flow in JavaScript
- Select and modify page elements using the DOM
- Respond to user actions with event listeners
- Work with arrays and objects to manage structured data
- Build small interactive browser programs from a blank page

WHAT WE'LL USE

Tools & Technologies

JavaScript (ES6+) VS Code Chrome DevTools CodePen

CURRICULUM

Module-by-Module Breakdown

MODULE	FOCUS
01 · JavaScript in the Browser	Where JS lives on a page, the console, linking scripts.
02 · Variables, Types & Operators	let / const, template literals, operators.

MODULE	FOCUS
03 · Control Flow & Functions	Conditionals, loops, function declarations and arrow functions.
04 · Arrays & Objects	Storing and manipulating structured data.
05 · The DOM	Selecting elements and changing content and styles dynamically.
06 · Events	Click, input and keyboard events; building real interactivity.
07 · Mini-Project Sprint	Combining the DOM, events and logic into one build.
08 · Debugging with DevTools & Capstone	Console methods, breakpoints, final project.

LEARNING BY BUILDING

Projects Students Will Build

- Interactive To-Do List — DOM manipulation, arrays and events.
- Digital Countdown Timer — Functions and live DOM updates.
- Rock-Paper-Scissors Game — Conditionals, randomization and the DOM.
- Browser Mini-Game — Timed events, scoring and interactivity.

BEYOND THE SYNTAX

Skills Developed

DOM manipulation Event-driven programming Browser debugging JavaScript fundamentals

PORTFOLIO OUTCOME

A live, deployed interactive JavaScript mini-app with a shareable link.

RECOMMENDED NEXT STEP

Web Development for Kids & Teens